LAB 04

Resistance in Series and Parallel & Voltage and Current Distribution (Dividers Law) in circuits

OBJECTIVE:

- Total resistance in a combination of resistors in series and parallel circuits.
- Distribution of the voltage and current to each resistor in a network.

EQUIPMENTS REQUIRED:

Sr. no	Name	Range	Quantity
1	Resistors		
2	DC variable voltage supply		
3	Digital Multimeter		
4	Others		

CIRCUIT DIAGRAM AND CALCULATION:

1. **Series resistance**

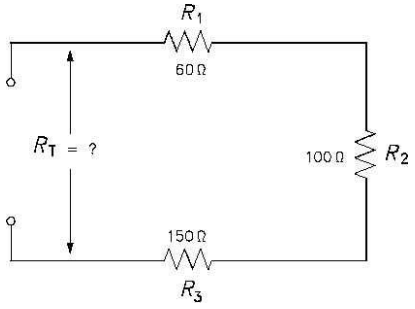


Figure 1. Resistors connected in series.

$$R_T = R_1 + R_2 + R_3$$

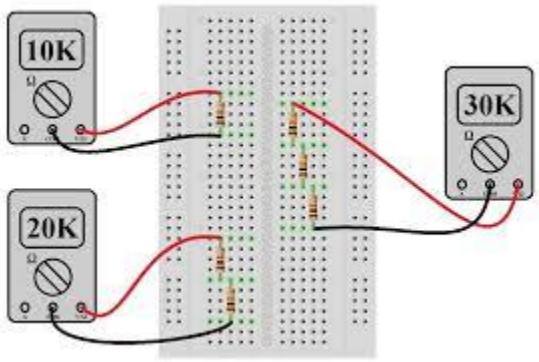


Figure 2. Snapshot of how resistors can be connected in series on breadboard.

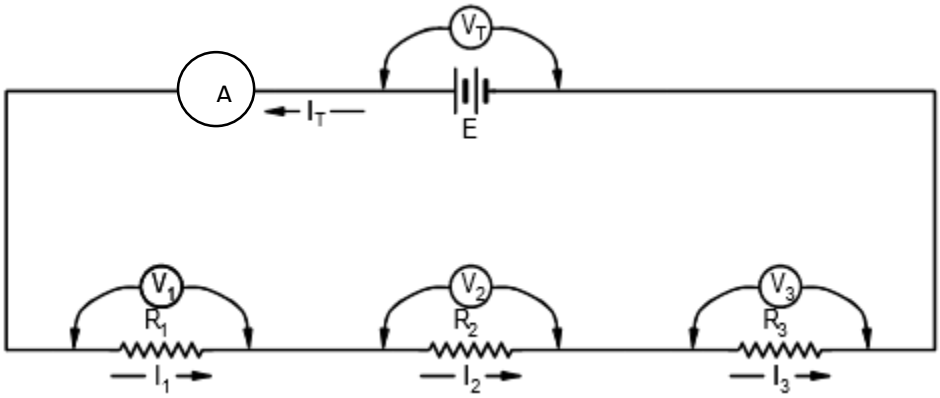
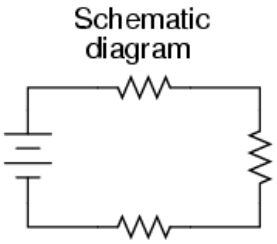
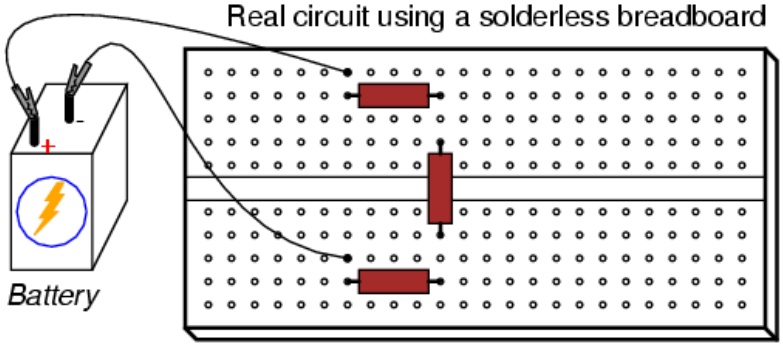


Figure 3. Voltage And Current Relation In A Resistive Series Circuit





The voltage drop across the battery V_T will be the total sum of the individual drops across each of the 3 resistors, and

$$\mathbf{V_T = V_1 + V_2 + V_3}$$

$$V_T = R_T I_T$$
$$V_1 = I_1 \times R_1$$
$$V_2 = I_2 \times R_2$$
$$V_3 = I_3 \times R_3$$

In this circuit the current supplied by the battery flows through each resistor, with the current in each resistor being the same. If the current supplied by the battery is I_T, the current in each resistor is I_1, I_2, and I_3, and they are all one and the same, then

$$I_T = I_1 = I_2 = I_3$$

PROCEDURE:

1. Identify the resistance of the resistors using the color code and write the value of each resistors in the table below.
2. Calculate the total resistance R_T and write in the table below.
3. Connect the resistors as shown in figure 1.
4. Using the multimeter in the resistance mode observe the resistance of the circuit at terminal R_T.
5. Write the observed value of the resistance in the table.

OBSERVATION TABLE:

Standard Color Code	Resistance Value From Color Code (Ω)	Observed Value From DMM (Ω)	Theoritical R_T (Ω)	Observed R_T (Ω)
R_1=				
R_2=				
R_3=				

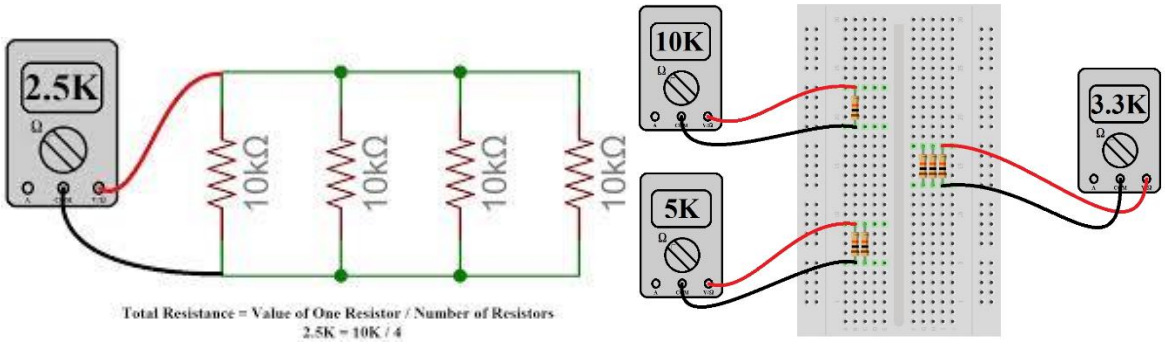
Procedure For Obtaining Voltage And Current Relation For Series Circuit:

1. Connect the circuit as shown in the figure.
2. Set the voltage of the voltmeter to 10 V.
3. Mark this voltage as V_T and note this voltage in the table below.
4. Using the DMM in the current mode observe the current I_T and write the value of current in the table.
5. Using DMM in the voltage mode observe the value for voltages across the resistors R_1 , R_2 , R_3 as V_1 , V_2 , V_3 respectively.
6. Using the values of R_1 , R_2 , R_3 and I_1 , I_2 , I_3 calculate the value for V_1 , V_2 , V_3 respectively.
7. Write these value in the table below.
8. Using the DMM in current mode write the values of current I_1 , I_2 , I_3 in the table.
9. Using values of the voltage across the resistors R_1 , R_2 , R_3 obtain the value of current I_1 , I_2 , I_3 .
10. Write these values in the table below.
11. Compare the observed and calculated values of the voltages and current to verify the results.

OBSERVATION TABLE:

Measured Voltage V_T(V)	Measured Voltage Across the resistors (V)	Calculated Voltage Across the resistors (V)	Calculated voltage V_T (V)	Measured current Across the resistors (I) mA	Calculated current Across the resistors (I)mA
V_T =	V_1 = V_2 = V_3 =	V_1 = V_2 = V_3 =	V_T =	I_1 = I_2 = I_3 =	I_1 = I_2 = I_3 =

2. Parallel resistance



Procedure For Measurement Of Parallel Resistance And Its Related Voltage And Current:

1. Using the resistance values R_1 , R_2 , R_3 calculate the total resistance using the formula for total resistance in parallel

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

2. Verify the value of the calculated resistance using the DMM.
3. Note these values in the table below.

OBSERVATION TABLE:

Standard Color Code	Resistance Value From Color Code (Ω)	Observed Value From DMM (Ω)	Theoritical R_T (Ω)	Observed R_T (Ω)
R_1=				
R_2=				
R_3=				

CIRCUIT DIAGRAM:

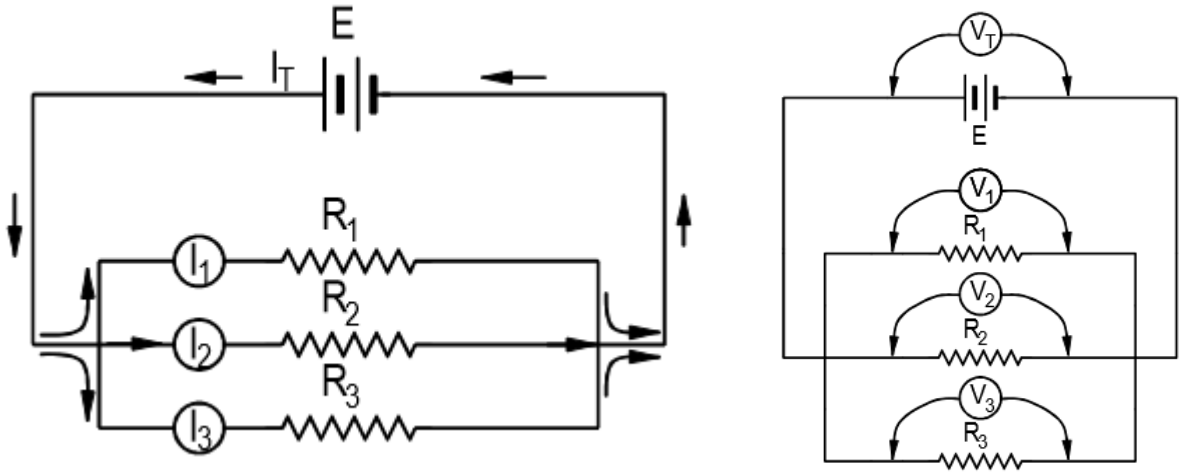


Figure 3. Three resistors R1, R2, and R3 connected in parallel showing the flow of current and potential differences all being equal.

The total current flowing in the circuit is the sum of the separate currents flowing through the resistors,

$$I_T = I_1 + I_2 + I_3$$

$$I_T = \frac{V_T}{R_T}.$$

$$I_1 = \frac{V_1}{R_1},$$

$$I_2 = \frac{V_2}{R_2},$$

$$I_3 = \frac{V_3}{R_3}.$$

The voltage drops across the battery and resistors are all equal, and as illustrated in Figure 3,

$$V_T = V_1 = V_2 = V_3.$$

Procedure For Obtaining Voltage And Current Relation For Parallel Circuit:

1. Using DMM measure the currents I_1 , I_2 , I_3 and . Also measure the value of V_T.
2. Wirte these value in the table below.
3. Using the value of resistors and the voltage across the it calculate the value of the currents I_1 , I_2 , I_3.
4. Note these value in the table below.
5. Use these value of current to calculate I_T.
6. Compare and verify the results.

OBSERVATION TABLE:

Calculated current I_T(A)	Calculated current Across the resisitors (A)	Measured current Across the resisitors (A)	Measured current I_T (A)	Calculated voltage Across the resisitors (V)	Measured voltage Across the resisitors (V)
I_T =	I_1 = I_2 = I_3 =	I_1 = I_2 = I_3 =	I_T =	V_1 = V_2 = V_3 =	V_1 = V_2 = V_3 =

Voltage Divider Rule:

The Voltage Divider Rule (VDR) states that the voltage across an element or across a series combination of elements in a series circuit is equal to the resistance of the element or series combination of elements divided by the total resistance of the series circuit and multiplied by the total impressed voltage:

$$V_n = \frac{R_n}{R_{total}} V_s$$

$V_n = voltage\ across\ the\ element,$

$R_n = resistance\ of\ the\ element$

$V_s = impressed\ voltage\ or\ supply\ voltage$

$$R_{total} = total\ resistance\ of\ the\ series\ circuit$$

CIRCUIT DIAGRAM:

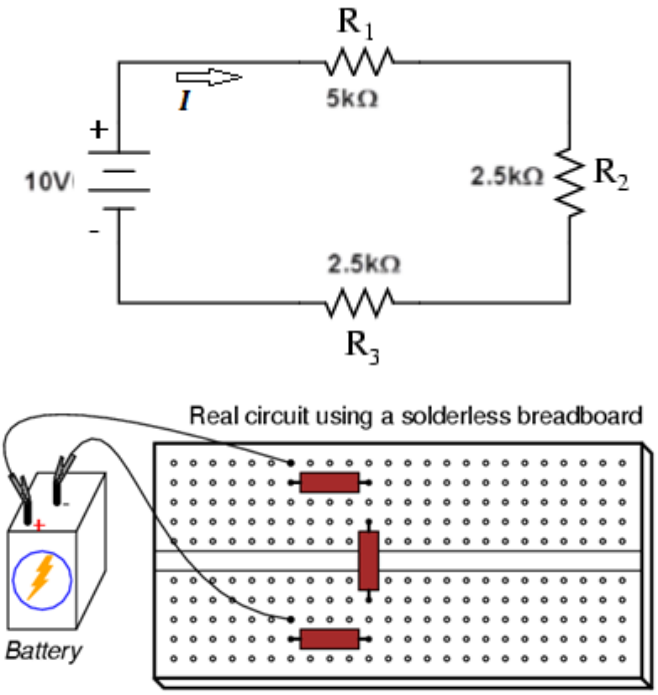


FIGURE 1. SEREIS CIRCUIT

CALCULATION:

$$I = \frac{V_s}{R_1 + R_2 + R_3} \quad ; \ current\ through\ the\ circuit$$

$$V_1 = \frac{R_1}{R_1 + R_2 + R_3} V_s \quad ; voltage\ across\ resistor\ R_1$$

$$V_2 = \frac{R_2}{R_1 + R_2 + R_3} V_s \quad ; voltage\ across\ resistor\ R_2$$

$$V_3 = \frac{R_3}{R_1 + R_2 + R_3} V_s \ ; \ voltage\ across\ resistor\ R_3$$

PROCEDURE FOR VOLTAGE DIVISION (VDR):

1. Connect the circuit shown in Fig. (1).
2. Take V_S = 10V, R_1=5KΩ, R_2=2.5KΩ and R_3=2.5KΩ.
3. Using ohm's law calculate the current "I" through the circuit in figure 1.
4. Practically verify this current using the digital multimeter.
5. Using the formula for the voltage divider calculate the voltage across each resistor.
6. Write the calculated voltage in the table below in the column for calculated value.

7. Measure the voltage across the resistors R_1, R_2 & R_3 using a digital multimeter in the DC voltage mode.
8. Write the value of the voltage across the resistors in the table below in the column for the observed value.
9. Write the remark from the result obtained from the observed and calculated values.

OBSERVATION TABLE:

Impressed / supply voltage (V)	Calculated Current (mA)	Observed Current (mA)	Calculated Value Of Voltage Across Resistors (V)	Observed Value Of Voltage Across Resistors (V)
$V_S =$	$I =$	$I =$	$V_1 =$	$V_1 =$
			$V_2 =$	$V_2 =$
			$V_3 =$	$V_3 =$
			$V_S = V_1 + V_2 + V_3 =$	$V_S = V_1 + V_2 + V_3 =$

The Current Divider Rule (CDR) states that the current through one of two parallel branches is equal to the resistance of the other branch divided by the sum of the resistances of the two parallel branches and multiplied by the total current entering the two parallel branches. That is,

$$I_n = \frac{R_{parallel}}{R_n} I_s$$

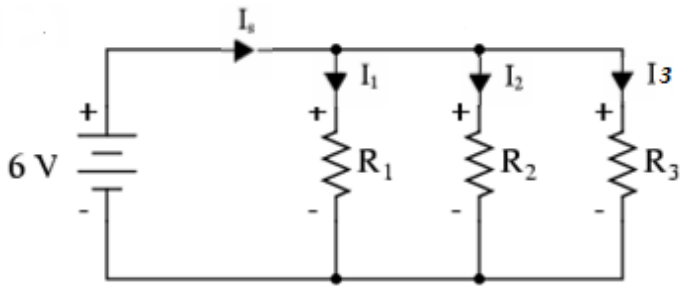
$I_n = current\ through\ one\ of\ the\ parallel\ branches$

$R_{parallel} = resistance\ of\ the\ other\ branch$

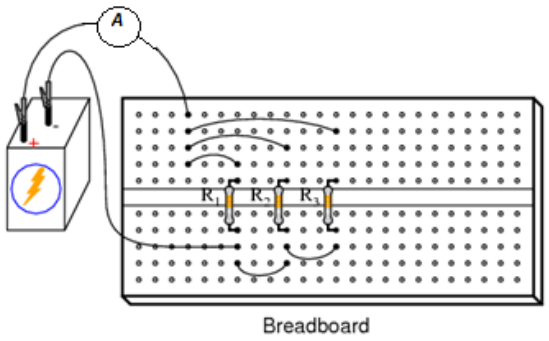
$R_n = sum\ of\ the\ resistance\ of\ the\ parallel\ branches$

$I_s = total\ current\ entering\ the\ branch$

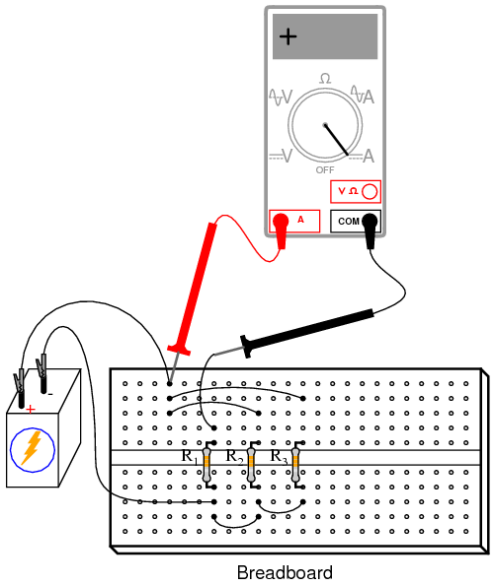
CIRCUIT DIAGRAM:



(a)



(b)



(c)

Figure 2. Current divider

CALCULATION:

$$I_s = \frac{V_s}{R_T}$$

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$I_1 = \frac{R_{(R_2//R_3)}}{R_1 + R_{(R_2//R_3)}} I_s$$

$$I_2 = \frac{R_3}{R_2 + R_3} (I_s - I_1)$$

$$I_3 = \frac{R_2}{R_3 + R_2} (I_s - I_1)$$

PROCEDURE FOR CURRENT DIVISION (CDR):

1. Connect the circuit as shown in Fig. (2a).
2. Take V_S = 6V, R_1=5KΩ, R_2=2.5KΩ and R_3=2.5KΩ.
3. Using ohm's law calculate the current I through the circuit in figure 2 (a).
4. Practically verify this current using the digital multimeter as shown in figure 2(b)
5. Using the formula for the current divider calculate the current flowing through each resistor.
6. Write the calculated current in the table below in the column for calculated value.
7. Measure the current through the resistors R_1, R_2 & R_3 using a digital multimeter in the DC current mode as in figure 2 (c).

8. Write the value of the current through the resistors in the table below in the column for the observed value.
9. Write the remark from the result obtained from the observed and calculated values.

OBSERVATION TABLE:

Impressed / supply voltage (V)	Calculated Current (mA)	Observed Current (mA)	Calculated Value Of Current Through Resistors (mA)	Observed Value Of Current Through Resistors (mA)
$V_S =$	$I_s =$	$I_s =$	$I_1 =$	$I_1 =$
			$I_2 =$	$I_2 =$
			$I_3 =$	$I_3 =$
			$I_s = I_1 + I_2 + I_3 =$	$I_s = I_1 + I_2 + I_3$

Conclusion:

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